

4.10.12

EE25BTECH11011 - Navya

Question:

The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

Solution:

The equations of the given lines are

$$\begin{pmatrix} 1 & 2 \end{pmatrix} \mathbf{x} = 10 \quad (1)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \mathbf{x} = -5 \quad (2)$$

$$\begin{pmatrix} 5 & 4 \end{pmatrix} \mathbf{x} = 0 \quad (3)$$

Forming a single matrix equation, we have

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 5 & 4 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 10 \\ -5 \\ 0 \end{pmatrix} \quad (4)$$

The augmented matrix is

$$\left(\begin{array}{cc|c} 1 & 2 & 10 \\ 2 & 1 & -5 \\ 5 & 4 & 0 \end{array} \right) \quad (5)$$

Applying row operations:

$$\left(\begin{array}{cc|c} 1 & 2 & 10 \\ 2 & 1 & -5 \\ 5 & 4 & 0 \end{array} \right) \xrightarrow{R_2 - 2R_1, R_3 - 5R_1} \left(\begin{array}{cc|c} 1 & 2 & 10 \\ 0 & -3 & -25 \\ 0 & -6 & -50 \end{array} \right) \quad (6)$$

$$\xrightarrow{R_2 / (-3)} \left(\begin{array}{cc|c} 1 & 2 & 10 \\ 0 & 1 & 25/3 \\ 0 & -6 & -50 \end{array} \right) \quad (7)$$

$$\xrightarrow{R_1 - 2R_2, R_3 + 6R_2} \left(\begin{array}{cc|c} 1 & 0 & -20/3 \\ 0 & 1 & 25/3 \\ 0 & 0 & 0 \end{array} \right) \quad (8)$$

Thus, the RREF gives the solution vector

$$\mathbf{x} = \begin{pmatrix} -\frac{20}{3} \\ \frac{25}{3} \end{pmatrix} \quad (9)$$

Hence, The straight line $5x + 4y = 0$ passes through the point of intersection of the straight lines $x + 2y - 10 = 0$ and $2x + y + 5 = 0$.

